

Examiner reports

Q1.

This question caused the candidates more difficulty. A relatively small number were able to produce completely correct solutions. The majority found part (a) difficult to deal with, but were often able to pick up marks with correct solutions for the more familiar part (b). Many of the candidates did not make coherent attempts at part (a) and produced confused working.

Q2.

The force diagram in part (a) was often drawn well, but the usual errors of extra forces and missing labels or arrows were seen. Part (b) was done very well, with the printed answer helping almost all of the candidates to gain full marks. The responses to part (c) were very varied, with some solutions excellent and some very poor. One fairly common error was to form an equation with a sign error, like $F - 5g \sin 40^\circ = 5 \times 0.8$. Other more serious errors included using the normal reaction force instead of the weight component parallel to the slope, and simply assuming that the friction was equal to 4. Part (d) of the question was found to be very difficult, with very few candidates giving acceptable answers. The majority of candidates stated that the coefficient of friction would increase.

Q3.

This question was answered well.

Q4.

Parts (a), (b) and (c) were answered well. In part (d), few candidates used the fact that the work done by the force was equal to the change in kinetic energy. Most used complicated methods which rarely gave an appropriate answer.

Q5.

Most candidates answered this question well, with only a few making algebraic or arithmetical errors.

Q6.

Virtually all candidates obtained $\frac{dv}{dt} = -0.05v$; only a few ignored the required step $m \frac{dv}{dt} = -0.05mv$. The equation $\int \frac{dv}{v} = - \int 0.05 \, dt$ was a necessary step which needed to be seen in part (b). Candidates knew roughly how to obtain $v = 20e^{-0.05t}$. Unfortunately, too often algebraic skills were not sufficient and the equation $\ln v = -0.05t + c$ regularly became $v = e^{-0.05t} + c$ before becoming $v = Ae^{-0.05t}$. This and similar errors were not condoned.

Q7.

The candidates found this question demanding, especially part (a). The few candidates who considered the forces acting on the caravan had few difficulties and often found the

correct force. Many candidates tried to consider the two bodies as a single particle and made little progress. Some candidates were able to do part (b), even if they had made little or no progress with part (a). As well as considering the forces on the wrong body, some candidates introduced a variety of sign errors into their equations. The real problem for the candidates in answering this question was that they did not recognise when to consider the two bodies separately and when to consider them as a single body.

Q8.

The first two parts of this question were done very well by the vast majority of candidates, but part (c) caused more difficulties. In part (a), the candidates almost always drew a graph with the correct shape, but made errors on the axes. The two most common errors were to use 21 instead of 12 on the time axis and to use an incorrect value on the velocity axis. In part (b), a very small number of candidates made arithmetic errors and a few also used incorrect values.

Part (c) caused more difficulties for the candidates. There were three main incorrect approaches. The first of these was to assume that the forces on the lift were in equilibrium and hence simply state that the tension was equal to the weight, giving their answer as 2940 N. Some candidates used an incorrect value for the acceleration, which they then included in a three term equation of motion which would have otherwise been correct. The most common approach was to take the acceleration as 2 m s^{-2} .

A few candidates used a three term equation of motion with the correct terms but an incorrect sign. This approach would produce a result such as $T = 2940 - 150 = 2790 \text{ N}$.

Q9.

Almost all candidates gained some marks on this question, but only a fairly small proportion gained full marks. In part (a), there were many good answers, but a number of candidates stated assumptions that were not about the string. Instead they produced comments about the block or the peg. Part (b) was generally done quite well, but there were some candidates who did not gain credit because they did not show enough working. For example, simply stating $6/4 = 1.5$ was not sufficient. There were, in contrast, some very good solutions that fully justified the final answer.

In part (c), the candidates who considered the forces on the 7 kg sphere did quite well. The main error that these candidates made was to use $T - 7g = 7a$, rather than $7g - T = 7a$ and obtained a tension of 79.1 N. Other candidates produced equations that involved the 13 kg block but could make little satisfactory progress because they did not know the friction force.

Part (d) was the most demanding part of the question. It required the candidates to find the friction force and then to use $F = \mu R$. Some candidates worked through these stages, but there were relatively few complete solutions. The most common errors were to assume that the friction was equal to the tension and simply apply $F = \mu R$, and to try to work with a combination of both bodies instead of just the block.

Q10.

In part (a) there were many good diagrams, but also a number that included extra forces, especially parallel to the slope. On some diagrams the arrows or labels were missing. Part (b) was done well and the vast majority of candidates did show enough working to gain all

of the marks. Some candidates clearly used the given answer to correct their working. Part (c) was more demanding. When finding the acceleration, the main errors were not dividing the resultant force by the mass; making a sign error to obtain $a = -0.171$; not considering both forces, to produce equations like $100a = 68.4$; and not resolving the weight correctly.

In the last part, there were some good explanations, but there were also a lot that were not acceptable. Some candidates wrote about the resistance being related to the acceleration rather than the speed or velocity. A few also thought that the resistance would depend on the slope.

Q11.

This question was generally done very well, particularly parts (a) and (b). In part (c), some candidates simply assumed that the tension would be equal to the weight. Some of these candidates gave the same answer for part (d) and gained the mark there. A few candidates performed the calculation $4410 - 45$ and obtained an incorrect value.

Q12.

Parts (a) and (b) of this question were done well by many candidates, but part (c) was more demanding. In part (a) most candidates formed two equations as requested, but a few only worked with one equation for the whole system. Those who had done part (a) correctly had few problems with part (b), while those with incorrect equations in part (a) obtained incorrect answers as a result. The main problem in part (c) was that many candidates assumed that the particles had moved 1 m each, instead of the 0.5 m that was needed for them to be 1 m apart. A few candidates took the acceleration to be 9.8 m s^{-2} in this part of the question.

Q13.

This question proved to be quite demanding for many candidates. In part (a), the main issue was that many candidates did not realise that they should consider the forces acting on the car. There were some who found two tensions, one based on the car and the other based on the truck. These solutions were often presented in such a way that they had to be treated as two attempts.

Many candidates were able to obtain the value of 600 in part (b), but some of the solutions did not contain any evidence of mechanics principles. If candidates were awarded full marks, they had to have demonstrated the consideration of the forces and an application of Newton's Second Law. Simply stating " $3000 - (800 + 1600) = 600$ " or similar was not considered sufficient.

Interestingly, there were many good answers to part (c), in some cases from candidates who scored no marks on (a) or (b). Some candidates were able to say that the tension would increase, but struggled to justify this.

Q14.

Although there were many correct force diagrams, there were also a significant number which included friction or other forces as well as the two required. A good number of candidates were able to find the acceleration when the slope was smooth, although there were a few candidates who did not show enough working, simply giving statements like " a

$\frac{1}{2}g = 4.9 \text{ m s}^{-2}$. In part (b), many candidates were able to find the acceleration successfully, but there were two common errors. The first was to assume that they could use $v = 5$ without any justification. The second was to simply divide the distance by the time. When finding the friction force, some candidates used $F = ma$ to obtain a friction force of 7.5 N. There were also a number of correct solutions. When finding the coefficient of friction, many candidates with incorrect values for the friction benefited from the follow through marks and gained full marks. There were however some cases where the normal reaction was incorrect either because no resolving had taken place or because the resolving had been incorrect.

In part (c), there were some good answers from candidates, but there were also quite a few which stated that the acceleration would be less, but did not give a good enough explanation for this.

Q15.

Most candidates completed this question successfully. A few used $\mathbf{v} = \mathbf{u} + \mathbf{a}t$ in part (b) and obtained $\mathbf{v} = (6 + 3t)\mathbf{i} + (30 - 6t^2)\mathbf{j}$, rather than the given answer. After they had done the integration of $\mathbf{a}$, a number of candidates 'invented' the $6\mathbf{i} + 30\mathbf{j}$ terms required.

Q16.

A significant number of candidates created $\frac{dv}{dt} = -\lambda v$ without proper justification. The equation $-\lambda mv = ma$ was a necessary step which needed to be seen. Even more knew roughly how to obtain $v = Ue^{-\lambda t}$, but too often algebraic skills were not sufficient and $\ln v = -\lambda t + \ln u$ regularly became $v = e^{-\lambda t} + U$ before becoming $v = Ue^{-\lambda t}$. This and similar errors were not condoned.

Q17.

Part (a) was often answered well but candidates appeared to find the requests complex and supplemented their answers with several lines of unnecessary working. Part (b)(i) proved popular and successful, but part (b)(ii) was very discriminating.

Q18.

Parts (a)(i), (ii), (v) and (b) proved routinely successful for well prepared and accurate candidates. Parts (a)(iii) and (iv) tested the understanding of the forces in the static situations and proved highly discriminating.

Q19.

This question required the use of a differential equation, but some candidates tried to use constant acceleration equations. There were a number of good, complete solutions. Some candidates were on the right track with their solutions, but made algebraic or integration errors. A few candidates omitted the negative sign when they started the question and ended up with a final answer which simply had a missing negative sign.

Q20.

In part (a), many of the candidates found the force correctly, but some did not go on to give the magnitude. A few candidates did not substitute $t = 0$. In part (b), the main problem was with the integration of the trigonometric terms. A particular problem was with the signs of the functions. Most candidates did attempt to find the constants of integration. A small number of candidates tried to use $\mathbf{v} = \mathbf{u} + \mathbf{a}t$.

Q21.

The responses to the modelling in parts (a) and (b) were very varied. Only a small number of candidates gained all three marks, but most gained at least one mark. Some candidates made statements that contradicted the information in the stem of the question, such as “the resistance force will be constant”. Part (c) tended to be done very well or very badly. There was some use of constant acceleration equations, but this was not very common. The biggest issue was the omission of a negative sign in the formation of the differential equation, which then continued throughout the solution. This error was quite common. Other problems were caused by incorrect algebraic manipulation, particularly with logarithms, or integration.

Q22.

It was pleasing that most candidates formed, as requested in part (a), equations of motion for the two particles. Pleasingly very few used the ‘whole approach instead of forming equations of motion for the two particles as requested. Most then proceeded to find the tension in the string. In part (b), rather than answer the question set, with a very small minority realising the necessity to consider the stone or the can in isolation.

Q23.

Diagrams in part (a) were clear and mostly correct, with occasional marks lost due to missing arrows or a missing or incorrect force, with the weight being given as ‘g’ as the most frequent. Parts (b) and parts (c) were also good, but few scored well in part (d), with one or even two forces missing from the equation of motion, the weight component being the most frequent omission.

Q24.

Again diagrams were mostly good, and parts (a)(ii) and (c) usually gained full marks. Part (b) was successfully completed by most candidates with the main error being the inclusion of the weight of P instead of the frictional force in the equation of motion of P. Pleasingly very few candidates used a ‘whole string’ approach instead of forming equations of motion for the two particles as requested. In part (d) the intended response focussed on the remaining force acting on each particle and the subsequent change in speed. Many candidates were successful here but others focussed on forces no longer present.

Q25.

Part (a) of this question was often done well. Many candidates seemed to take the question in their stride, but others made limited progress or did not know how to approach the question. There were some who lost accuracy marks. The four main causes of this were;

- Omission of the negative sign in the differential equation,

- Algebraic errors,
- Incorrect integration,
- Not finding the value of the constant of integration.

Part (b) was done very well, with many candidates gaining the available mark without doing part (a).

Q26.

The candidates generally found part (a) of this question very difficult. Some of the more able candidates found an expression for F in terms of t and then applied Newton's second law. A few effectively reversed this process, by changing the graph into one for acceleration and then finding the equation of the line. Due to the fact that the answer was given, there were a large number of partial or very unconvincing arguments. Some candidates simply omitted this part of the question and went straight on to part (b).

There were many good responses to part (b), but a common error was to ignore the constant of integration. Candidates do need to include a constant of integration and find its value, even if it is zero, in order to gain full marks.

There was a similar problem in part (c), with many candidates again ignoring the constant of integration. Those candidates who included limits of integration, experienced less difficulty, because they wrote down the values of 0 and 200 and substituted both of them.

A few candidates used constant acceleration equations, but the number doing this was fairly small.

Part (d) produced some very mixed responses. There were some very good explanations, but some were very confused. Some candidates wrote about the acceleration instead of the velocity. In some cases the candidates seemed to have some idea of how to respond, but did not express this clearly. Quite a few candidates said that the constant would change, which seemed to imply that they were thinking about the intercept of the graph, but this did not lead to any marks as the velocity does not contain a constant term.

Q27.

Most candidates did well on this question; almost all were able to obtain the acceleration without difficulty. Finding the distance was also done well, but a few candidates made arithmetic errors. There was also a small number who assumed a constant velocity. Part (b) caused a few more problems. Some candidates had trouble with the units and did not convert tonnes to kg. A few also made sign errors, while some others simply calculated the product of the mass and acceleration.

Q28.

There was a good number of correct force diagrams. Some candidates added one or both of the components of the tension to their force diagram. In some cases these appeared as if they were additional forces. Parts (b) and (c) were done well by most candidates who seemed to have benefitted from the printed answers. There were some candidates who did not gain full marks, because they did not show sufficient working. In part (d), many candidates obtained the correct value. Some candidates used the two printed answers as expected, if they had not been able to obtain the two values for themselves. There were some very good answers to part (e), but there were also quite a few confused ones as well, with many candidates finding it hard to give a good answer.

Q29.

Candidates found part (a) quite straightforward, although some did omit g from their calculations. Part (b) was done well by some candidates, but some of the weaker candidates used the weight of the 8kg particle rather than the tension when forming an equation to find its acceleration.

Q30.

There was a good number of correct force diagrams. Some candidates added one or both of the components of the tension to their force diagram. In some cases these appeared as if they were additional forces. Parts (b) and (c) were done well by most candidates who seemed to have benefitted from the printed answers as expected. There were some candidates who did not gain full marks, because they did not show sufficient working. In part (d), many candidates obtained the correct value. Some candidates used the two printed answers as expected, if they had not been able to obtain the two values for themselves. There were some very good answers to part (e), but there were also quite a few confused ones as well, with many candidates finding it hard to give a good answer.

Q31.

In part (a), a number of candidates made assumptions about the particles instead of the pulley. Part (b) was generally done very well, but some candidates ignored the instruction to form equations of motion for each particle; these candidates were penalised. Most candidates were able to find the tension.

Q32.

Generally this question was done very well. An error that was observed several times in part (a) was to find the area above the graph, rather than the area below the graph. Part (b) was done very well. Part (c) was also done well, but some candidates rounded their final answer to two significant figures.

Q33.

Generally the candidates did well on this question. In part (a) there were a relatively small number of candidates who ignored the instruction to form an equation of motion for each particle. Some candidates used a method that produced -2.8 ms^{-2} . This approach was acceptable for part (a), but these candidates then used 2.8 in their equation to find the tension and obtained an incorrect value. In part (c), there were many very good responses. Some candidates failed to double their answer and lost the final mark. A few candidates worked with the acceleration due to gravity, rather than the acceleration that they had calculated in part (a).

Q34.

A variety of force diagrams were seen on the candidates' scripts. Some were very good, but errors included the omission of forces and showing the friction force acting up the slope. In part (b) the candidates were clearly helped by the printed answer and there were many correct responses. In some cases the candidates did not show enough working to gain full marks. Part (c) proved to be very challenging, with only a very small number of candidates getting the correct coefficient of friction. The main problem was that the candidates did not include all of the terms that were needed in their equations of motion.

Q35.

Good candidates tended to score well on this question while weaker candidates would only gain a few marks. Part (a) was often answered well, but candidates often missed the key assumption that was being made and stated assumptions that were not appropriate. In part (b) the most common error was to take the mass as 200 kg rather than 200 grams. When answering part (c), some candidates stated that the initial acceleration would be 8 ms^{-2} and some said that it would be zero. In the explanation, the quality of the expression was often poor and lacked clarity. Also some candidates did not explicitly state what would happen to the acceleration, but described what would happen to the velocity instead.



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